🧠 Part A: Conceptual Understanding (Theory)

1. What is Logistic Regression and why is it suitable for classification?

Logistic Regression is a supervised machine learning algorithm used to predict categorical outcomes, mainly binary classes (0 or 1). It uses the sigmoid function to convert linear predictions into probabilities between 0 and 1, making it ideal for classification tasks.

2. Explain classification performance metrics and why accuracy alone is insufficient.

Classification metrics include Accuracy, Precision, Recall, F1-Score, ROC-AUC, and Confusion Matrix. Accuracy alone can be misleading for imbalanced datasets because a model may predict the majority class correctly while completely missing the minority class.

3. Define Type-I Error and Type-II Error in the context of risk prediction.

- **Type-I Error (False Positive):** Predicting a risk when there is actually no risk.
- **Type-II Error (False Negative):** Failing to predict a real risk that actually exists. In many applications, Type-II errors can have more serious consequences.

4. Explain Precision, Recall, F1-Score, TPR, and FPR.

- **Precision:** Percentage of predicted positives that are actually positive.
- **Recall (TPR):** Percentage of actual positives correctly identified.
- **F1-Score:** Harmonic mean of Precision and Recall, balancing both metrics.
- **FPR:** Percentage of actual negatives incorrectly classified as positive.

5. What is AUC-ROC and how does it help in evaluating classifiers?

The ROC curve plots the True Positive Rate against the False Positive Rate at different classification thresholds. The **AUC (Area Under the Curve)** measures the model's ability to distinguish between classes—higher AUC indicates better classification performance.

6. Why does imbalanced data create problems in classification models?

In imbalanced datasets, one class has significantly more samples than the other. Models tend to favor the majority class, resulting in poor detection of the minority class and misleadingly high accuracy, making metrics like Precision, Recall, and F1-Score more appropriate.